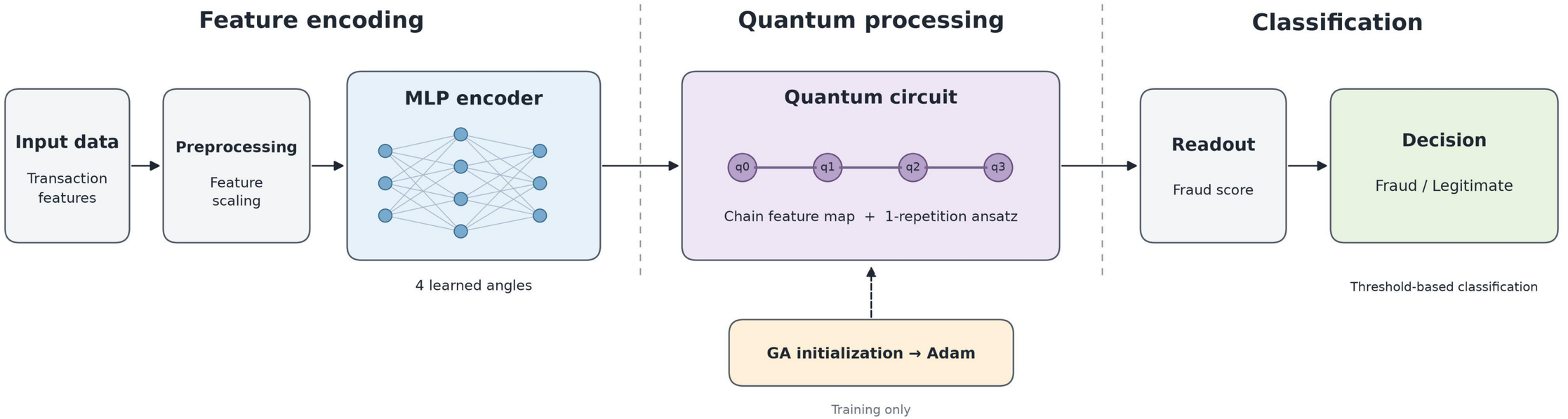
Quantum Prompting

OBIECTIVE

1. O reprezentare mai bogata a datelor de intrare;
2. Reducerea numarul de shoturi;
3. Antrenament mai avantajos computational;
4. Degradarea sub zgomot depolarizant simulat

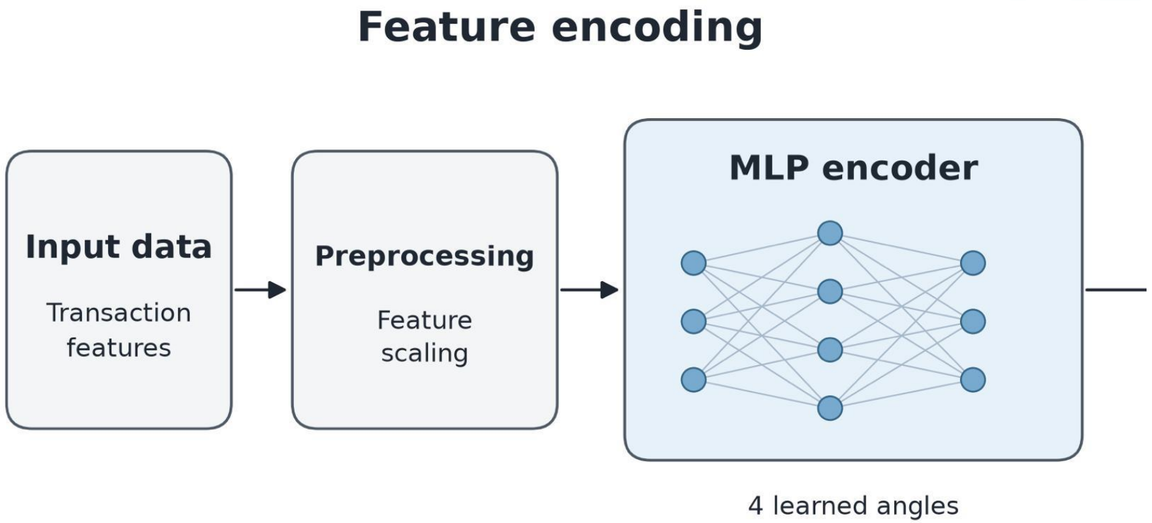
CONTRIBUȚIA PROPUSA

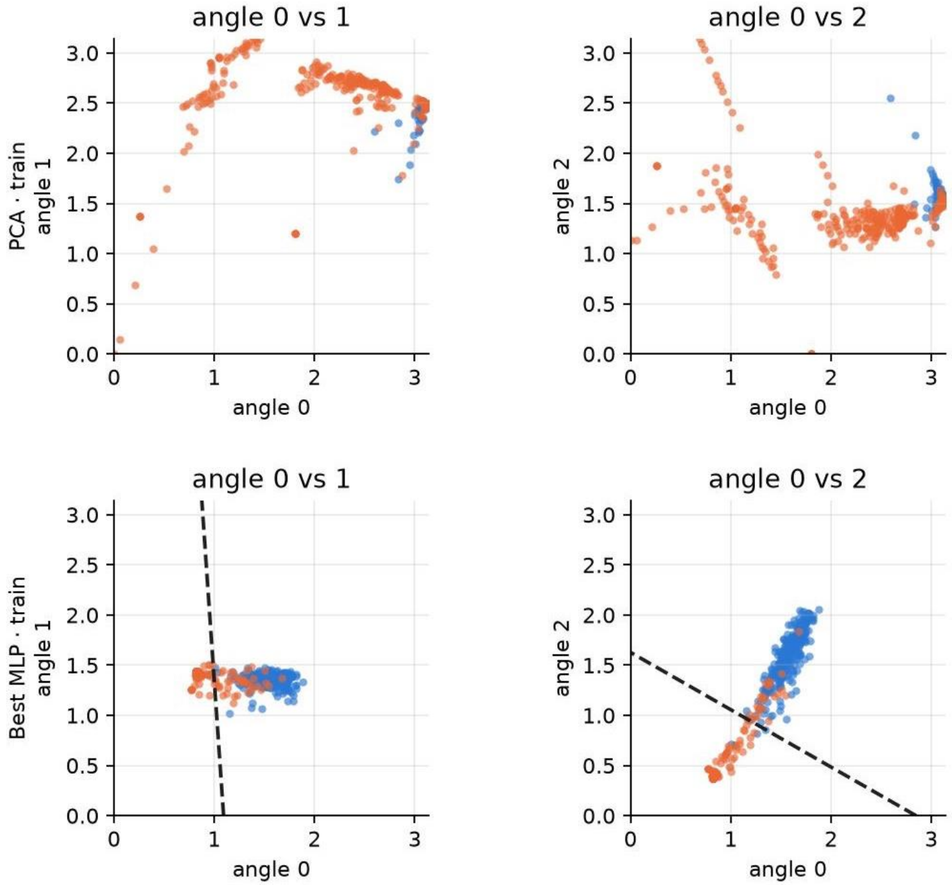
MLP + GA + VQC



CONTRIBUȚIA PROPUSA

MLP





- **Separă frauda de tranzacțiile legitime.**
- **Înlocuiește unghiurile din PCA nesupravegheat.**
- **Dă circuitului semnalul de clasă.**

CONTRIBUȚIA PROPUSA
Genetic Algorithm (GA)



CONTRIBUȚIA PROPUSA VQC

All-to-all

3 straturi ansatz

32 parametri

93 porți

Chain

1 strat ansatz

16 parametri

53 porți

Mai puține porți → cost cuantic redus

CONFIGURAȚIA EXPERIMENTELOR ȘTIINȚIFICE

1. Aceleași date

Tranzacții bancare

60% antrenare
20% validare
20% test

Clase echilibrate doar la antrenare

Împărțire fixă, stratificată

2. Cele 6 modele

Baseline
PCA + circuit cuantic

Head
Rețea clasică, fără circuit

Two phase
Unghiuri învățate → circuit

Two phase + GA
Inițializare genetică

Baseline Chained
Circuit mai simplu, în lanț

Two phase + GA + Chained
Unghiuri + GA + circuit în lanț

3. Sweep de parametri

Încercăm combinații aleatorii

Rețea clasică
Dimensiune, epoci, rată de învățare și regularizare

Two phase + GA + Chained
Populație, generații, mutație, elite + epoci și rată de învățare VQC

4. Antrenare și comparație pe test

Fără zgomot la antrenare
Repetări cu seed-uri diferite

Test cu și fără zgomot cuantic
p = 0; 0,001; 0,01; 0,05 • Head: fără zgomot

Performanță + robustețe + cost
AUPRC și număr de execuții de circuit

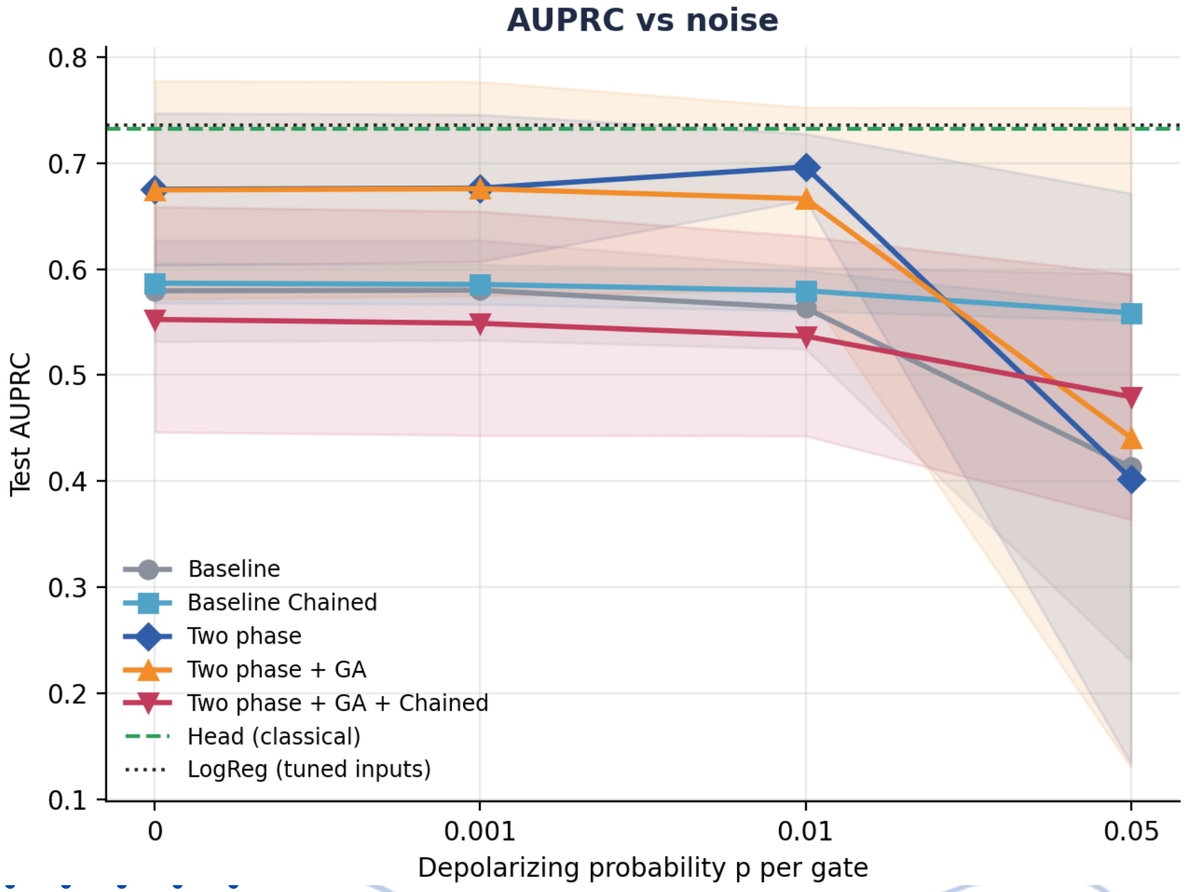
REZULTATE

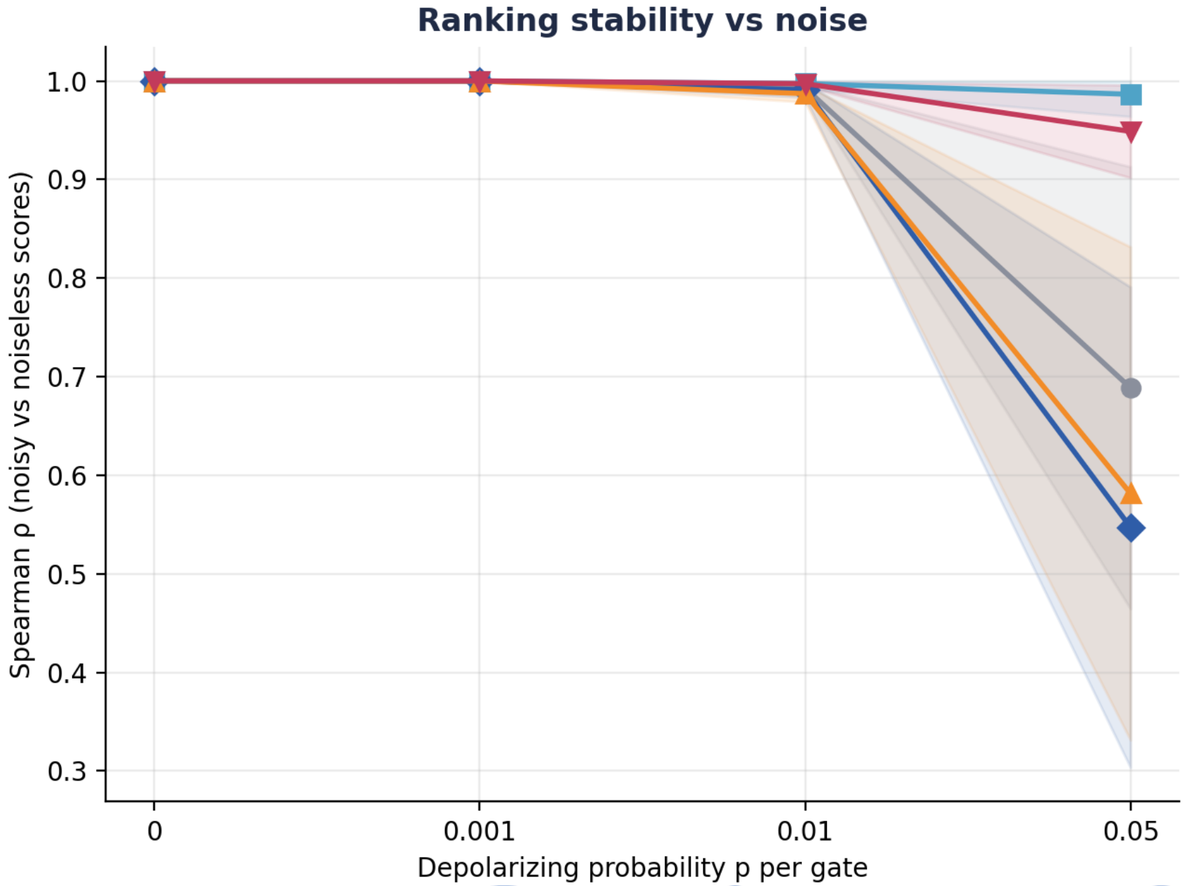
Test performance under depolarizing noise

Mean ± std over 5 seeds · common split (test keeps the 0.17% fraud rate) · threshold chosen on noisy validation scores

Model	Params	AUPRC p = 0	AUPRC p = 0.01	AUPRC p = 0.05	Spearman ρ p = 0.05	F1 p = 0.05	Train+val exec. (M)
Baseline	32	0.580 ± 0.048	0.563 ± 0.039	0.413 ± 0.182	0.69 ± 0.22	0.538 ± 0.189	2.86
Baseline Chained	16	0.587 ± 0.019	0.580 ± 0.019	**0.559 ± 0.007**	**0.99 ± 0.02**	**0.638 ± 0.011**	2.29
Two phase	32	**0.675 ± 0.072**	**0.696 ± 0.031**	0.402 ± 0.269	0.55 ± 0.24	0.532 ± 0.313	2.86
Two phase + GA	32	**0.675 ± 0.103**	0.666 ± 0.086	0.441 ± 0.311	0.58 ± 0.25	0.557 ± 0.316	1.46
Two phase + GA + Chained	16	0.553 ± 0.106	0.537 ± 0.094	0.480 ± 0.116	0.95 ± 0.05	0.488 ± 0.312	**0.73**
Head (classical)	5	0.733 ± 0.029	—	—	—	0.794 ± 0.014*	0

REZULTATE


AUPRC vs noise
Test AUPRC
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.001
0.01
0.05
Depolarizing probability p per gate
Baseline
Baseline Chained
Two phase
Two phase + GA
Two phase + GA + Chained
Head (classical)
LogReg (tuned inputs)


Ranking stability vs noise
Spearman ρ (noisy vs noiseless scores)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
0.001
0.01
0.05
Depolarizing probability p per gate

CONCLUZII

1. Circuit simplificat -> mai robust la zgomot;
2. Antrenarea in 2 faze ajuta;
3. Validare mai rara -> cost mai mic;

Ideea principala: Un VQC eficient cere echilibru

intre performanta, robustete si cost

LIMITELE CERCETĂRII

LIMITARI:

- **Zgomotul e un simulator, iar antrenarea e curată.**
- Fix 2 elite pe 6 candidati

VA MULȚUMIM!

ANEXE

Cele 6 modele

Appendix · full results under noise (page 1/2)

Mean ± std over 5 seeds · test set · threshold = max F1 on validation at the same noise level · p = depolarizing probability per gate

Model	p	AUPRC	AUROC	Precision	Recall	F1	MCC	Spearman ρ	Threshold
Baseline	0	0.580 ± 0.048	0.967 ± 0.004	0.678 ± 0.066	0.616 ± 0.031	0.644 ± 0.033	0.645 ± 0.034	1.000	0.617 ± 0.041
	0.001	0.580 ± 0.047	0.967 ± 0.004	0.674 ± 0.067	0.616 ± 0.031	0.642 ± 0.033	0.643 ± 0.034	1.000 ± 0.000	0.603 ± 0.037
	0.01	0.563 ± 0.039	0.967 ± 0.003	0.695 ± 0.059	0.598 ± 0.039	0.641 ± 0.021	0.643 ± 0.022	0.993 ± 0.004	0.544 ± 0.015
	0.05	0.413 ± 0.182	0.957 ± 0.017	0.495 ± 0.220	0.631 ± 0.127	0.538 ± 0.189	0.549 ± 0.176	0.689 ± 0.224	0.500 ± 0.001
Baseline Chained	0	0.587 ± 0.019	0.933 ± 0.006	0.614 ± 0.024	0.682 ± 0.020	0.646 ± 0.006	0.646 ± 0.005	1.000	0.563 ± 0.033
	0.001	0.586 ± 0.019	0.934 ± 0.006	0.615 ± 0.023	0.682 ± 0.020	0.646 ± 0.005	0.647 ± 0.004	1.000 ± 0.000	0.562 ± 0.027
	0.01	0.580 ± 0.019	0.940 ± 0.009	0.610 ± 0.033	0.680 ± 0.023	0.642 ± 0.009	0.643 ± 0.008	0.997 ± 0.005	0.546 ± 0.025
	0.05	0.559 ± 0.007	0.960 ± 0.001	0.594 ± 0.023	0.690 ± 0.015	0.638 ± 0.011	0.639 ± 0.010	0.986 ± 0.023	0.508 ± 0.007
Two phase	0	0.675 ± 0.072	0.967 ± 0.006	0.757 ± 0.035	0.786 ± 0.047	0.770 ± 0.026	0.770 ± 0.026	1.000	0.911 ± 0.042
	0.001	0.676 ± 0.069	0.966 ± 0.006	0.758 ± 0.034	0.788 ± 0.048	0.772 ± 0.026	0.772 ± 0.027	1.000 ± 0.000	0.870 ± 0.038
	0.01	0.696 ± 0.031	0.965 ± 0.006	0.759 ± 0.035	0.792 ± 0.040	0.774 ± 0.026	0.775 ± 0.026	0.991 ± 0.008	0.643 ± 0.022
	0.05	0.402 ± 0.269	0.884 ± 0.121	0.499 ± 0.312	0.755 ± 0.107	0.532 ± 0.313	0.544 ± 0.297	0.547 ± 0.244	0.503 ± 0.002
Two phase + GA	0	0.675 ± 0.103	0.966 ± 0.012	0.764 ± 0.043	0.790 ± 0.040	0.776 ± 0.024	0.776 ± 0.024	1.000	0.905 ± 0.033
	0.001	0.676 ± 0.101	0.966 ± 0.012	0.764 ± 0.043	0.790 ± 0.040	0.776 ± 0.024	0.776 ± 0.024	1.000 ± 0.000	0.864 ± 0.030
	0.01	0.666 ± 0.086	0.966 ± 0.011	0.741 ± 0.046	0.798 ± 0.038	0.767 ± 0.024	0.768 ± 0.024	0.987 ± 0.009	0.640 ± 0.016
	0.05	0.441 ± 0.311	0.940 ± 0.033	0.541 ± 0.338	0.773 ± 0.099	0.557 ± 0.316	0.578 ± 0.287	0.582 ± 0.250	0.503 ± 0.001
Head (classical)	—	0.733 ± 0.029	0.976 ± 0.002	0.781 ± 0.039	0.810 ± 0.026	0.794 ± 0.014	0.795 ± 0.013	—	0.716 ± 0.084
Two phase + Muon	0	0.714 ± 0.018	0.964 ± 0.006	0.756 ± 0.029	0.802 ± 0.036	0.777 ± 0.019	0.778 ± 0.019	1.000	0.874 ± 0.073
	0.001	0.717 ± 0.015	0.964 ± 0.006	0.756 ± 0.029	0.802 ± 0.036	0.777 ± 0.019	0.778 ± 0.019	1.000 ± 0.000	0.836 ± 0.066
	0.01	0.703 ± 0.036	0.962 ± 0.007	0.769 ± 0.051	0.778 ± 0.068	0.771 ± 0.036	0.772 ± 0.036	0.990 ± 0.004	0.633 ± 0.025
	0.05	0.165 ± 0.147	0.892 ± 0.099	0.277 ± 0.233	0.718 ± 0.151	0.333 ± 0.262	0.369 ± 0.236	0.579 ± 0.192	0.502 ± 0.002
Two phase Chained	0	0.532 ± 0.236	0.953 ± 0.020	0.634 ± 0.251	0.792 ± 0.019	0.675 ± 0.195	0.692 ± 0.163	1.000	0.838 ± 0.117
	0.001	0.530 ± 0.235	0.953 ± 0.020	0.625 ± 0.252	0.796 ± 0.023	0.671 ± 0.195	0.688 ± 0.163	1.000 ± 0.000	0.825 ± 0.114
	0.01	0.473 ± 0.210	0.953 ± 0.021	0.580 ± 0.242	0.751 ± 0.097	0.627 ± 0.191	0.644 ± 0.164	0.996 ± 0.004	0.737 ± 0.095
	0.05	0.306 ± 0.289	0.948 ± 0.023	0.407 ± 0.351	0.796 ± 0.054	0.465 ± 0.290	0.520 ± 0.239	0.922 ± 0.070	0.555 ± 0.034

Spearman ρ = rank correlation between noisy and noiseless test scores (1.000 at p = 0 by definition). Head = classical, no noise applied. LogReg = logistic regression on the same inputs.

ANEXE

Alte experimente rulate

Appendix · full results under noise (page 2/2)

Mean ± std over 5 seeds · test set · threshold = max F1 on validation at the same noise level · p = depolarizing probability per gate

Model	p	AUPRC	AUROC	Precision	Recall	F1	MCC	Spearman ρ	Threshold
Two phase + GA + Chained	0	0.553 ± 0.106	0.946 ± 0.023	0.569 ± 0.315	0.798 ± 0.035	0.603 ± 0.304	0.625 ± 0.261	1.000	0.870 ± 0.078
	0.001	0.549 ± 0.106	0.946 ± 0.023	0.592 ± 0.318	0.759 ± 0.093	0.599 ± 0.304	0.619 ± 0.261	1.000 ± 0.000	0.861 ± 0.079
	0.01	0.537 ± 0.094	0.947 ± 0.024	0.564 ± 0.311	0.735 ± 0.152	0.570 ± 0.300	0.592 ± 0.258	0.997 ± 0.002	0.774 ± 0.066
	0.05	0.480 ± 0.116	0.946 ± 0.026	0.434 ± 0.343	0.816 ± 0.052	0.488 ± 0.312	0.535 ± 0.262	0.949 ± 0.047	0.577 ± 0.025
Lightcone (Adam)	0	0.616 ± 0.185	0.962 ± 0.011	0.711 ± 0.135	0.776 ± 0.044	0.734 ± 0.079	0.738 ± 0.070	1.000	1.218 ± 0.294
	0.001	0.616 ± 0.185	0.962 ± 0.011	0.712 ± 0.133	0.776 ± 0.044	0.735 ± 0.077	0.739 ± 0.069	1.000 ± 0.000	1.045 ± 0.258
	0.01	0.607 ± 0.185	0.964 ± 0.009	0.703 ± 0.143	0.784 ± 0.047	0.731 ± 0.080	0.737 ± 0.071	0.980 ± 0.010	0.120 ± 0.107
	0.05	0.141 ± 0.166	0.952 ± 0.009	0.231 ± 0.231	0.692 ± 0.194	0.276 ± 0.156	0.347 ± 0.130	0.632 ± 0.183	-0.390 ± 0.126
Lightcone (AdamW)	0	0.620 ± 0.171	0.964 ± 0.007	0.720 ± 0.124	0.778 ± 0.036	0.741 ± 0.072	0.744 ± 0.065	1.000	1.225 ± 0.318
	0.001	0.618 ± 0.172	0.964 ± 0.007	0.719 ± 0.125	0.778 ± 0.036	0.741 ± 0.073	0.744 ± 0.066	1.000 ± 0.000	1.046 ± 0.285
	0.01	0.611 ± 0.178	0.966 ± 0.006	0.701 ± 0.130	0.786 ± 0.044	0.733 ± 0.072	0.738 ± 0.064	0.983 ± 0.012	0.095 ± 0.176
	0.05	0.148 ± 0.118	0.954 ± 0.008	0.248 ± 0.163	0.704 ± 0.172	0.328 ± 0.148	0.385 ± 0.119	0.668 ± 0.185	-0.415 ± 0.202
Lightcone (Muon)	0	0.616 ± 0.114	0.956 ± 0.007	0.738 ± 0.060	0.706 ± 0.131	0.715 ± 0.075	0.718 ± 0.073	1.000	1.222 ± 0.260
	0.001	0.616 ± 0.115	0.957 ± 0.007	0.739 ± 0.058	0.704 ± 0.139	0.714 ± 0.081	0.717 ± 0.077	1.000 ± 0.000	1.034 ± 0.245
	0.01	0.543 ± 0.190	0.958 ± 0.005	0.681 ± 0.124	0.651 ± 0.208	0.654 ± 0.156	0.659 ± 0.150	0.979 ± 0.011	0.088 ± 0.237
	0.05	0.039 ± 0.026	0.927 ± 0.031	0.074 ± 0.056	0.678 ± 0.190	0.131 ± 0.096	0.211 ± 0.125	0.630 ± 0.216	-0.478 ± 0.245
Two phase (untuned)	0	0.597 ± 0.109	0.951 ± 0.017	0.731 ± 0.086	0.702 ± 0.064	0.715 ± 0.070	0.715 ± 0.071	1.000	0.859 ± 0.047
	0.001	0.591 ± 0.123	0.951 ± 0.017	0.727 ± 0.097	0.720 ± 0.054	0.720 ± 0.058	0.722 ± 0.057	1.000 ± 0.000	0.821 ± 0.039
	0.01	0.563 ± 0.194	0.951 ± 0.016	0.697 ± 0.188	0.724 ± 0.055	0.702 ± 0.133	0.706 ± 0.126	0.984 ± 0.013	0.623 ± 0.012
	0.05	0.167 ± 0.274	0.902 ± 0.028	0.222 ± 0.331	0.590 ± 0.223	0.241 ± 0.322	0.275 ± 0.297	0.594 ± 0.240	0.502 ± 0.001
Two phase + Muon (untuned)	0	0.636 ± 0.055	0.938 ± 0.023	0.755 ± 0.057	0.724 ± 0.059	0.738 ± 0.047	0.738 ± 0.047	1.000	0.810 ± 0.025
	0.001	0.638 ± 0.057	0.938 ± 0.023	0.748 ± 0.058	0.749 ± 0.029	0.747 ± 0.032	0.748 ± 0.032	1.000 ± 0.000	0.775 ± 0.023
	0.01	0.579 ± 0.101	0.938 ± 0.023	0.702 ± 0.079	0.741 ± 0.042	0.717 ± 0.030	0.719 ± 0.029	0.984 ± 0.006	0.607 ± 0.009
	0.05	0.222 ± 0.273	0.806 ± 0.260	0.275 ± 0.336	0.673 ± 0.186	0.297 ± 0.324	0.326 ± 0.302	0.616 ± 0.147	0.502 ± 0.001
LogReg · inputs of: Two phase, Two phase + Muon	0	0.680 ± 0.046	0.975 ± 0.002	0.718 ± 0.026	0.806 ± 0.041	0.759 ± 0.023	0.760 ± 0.024	—	0.990 ± 0.010
LogReg · inputs of: Baseline, Baseline Chained	0	0.703 ± 0.010	0.976 ± 0.003	0.776 ± 0.029	0.808 ± 0.013	0.791 ± 0.019	0.791 ± 0.018	—	0.791 ± 0.030
LogReg · inputs of: Two phase + GA + Chained, Two phase Chained	0	0.704 ± 0.025	0.974 ± 0.003	0.792 ± 0.025	0.790 ± 0.043	0.790 ± 0.029	0.790 ± 0.028	—	0.984 ± 0.010
LogReg · inputs of: Head, Lightcone, Two phase...	0	0.735 ± 0.033	0.976 ± 0.002	0.780 ± 0.041	0.806 ± 0.030	0.792 ± 0.017	0.792 ± 0.017	—	0.984 ± 0.008

Spearman ρ = rank correlation between noisy and noiseless test scores (1.000 at p = 0 by definition). Head = classical, no noise applied. LogReg = logistic regression on the same inputs.